AME 5101 - Applied Linear Algebra

Assignment 3B

Maximum marks: 10 Duration: 45 minutes

The Assignment Question.

Q1. [CO3-PO3-BL3] **10 Marks**

This question uses the network of Figure 1 in the notes *Backpropagation with Cross-Entropy Loss*: two inputs, two hidden layers of three ReLU units each, and three output units (the logits). Work it on the same lines as the worked computation in Part II of the notes: the sections *Data*, *Forward pass and loss* and *Gradient at the logits*. Show every product and sum, and keep four decimal places.

The data are

$$\mathbf{W}^{(1)} = \begin{pmatrix} 0.4 & -0.2 \\ -0.5 & 0.3 \\ 0.2 & 0.6 \end{pmatrix}, \qquad \mathbf{b}^{(1)} = \begin{pmatrix} 0 \\ 0.1 \\ 0 \end{pmatrix},$$

$$\mathbf{W}^{(2)} = \begin{pmatrix} 0.3 & 0.5 & -0.2 \\ -0.4 & 0.2 & 0.5 \\ 0.6 & -0.3 & 0.1 \end{pmatrix}, \qquad \mathbf{W}^{(3)} = \begin{pmatrix} 0.2 & 0.4 & -0.3 \\ 0.5 & -0.1 & 0.2 \\ -0.2 & 0.3 & 0.4 \end{pmatrix},$$

with $\mathbf{b}^{(2)} = \mathbf{b}^{(3)} = \mathbf{0}$, input $\mathbf{x} = (2,\ 1)^{\top}$ and label $c = 3$.

(a) [3] Compute $\mathbf{z}^{(1)} = \mathbf{W}^{(1)}\mathbf{x} + \mathbf{b}^{(1)}$ and $\mathbf{a}^{(1)} = \mathrm{ReLU}(\mathbf{z}^{(1)})$. Which unit of Layer 1 is inactive? Does the bias change which units are active here?

(b) [2] Compute $\mathbf{z}^{(2)}$ and $\mathbf{a}^{(2)}$. Which column of $\mathbf{W}^{(2)}$ has no effect on $\mathbf{z}^{(2)}$, and why?

(c) [1] Compute the logits $\mathbf{z}^{(3)}$. Which column of $\mathbf{W}^{(3)}$ has no effect here?

(d) [2] Compute $\mathbf{q} = \mathrm{softmax}(\mathbf{z}^{(3)})$ and check that its entries add to 1. Compute the loss $L = -\ln q_c$. Is L above or below $\ln 3 \approx 1.0986$, and why?

(e) [2] Write down the target $\mathbf{y}$ and compute the gradient at the logits, $\boldsymbol{\delta}^{(3)} = \mathbf{q} - \mathbf{y}$. Check that its entries sum to zero. Which logit would a gradient-descent step raise, and which would it lower?